Question	Answer	Marks	Partial Marks
3(d)	Tangent ruled at $x = 2$	**B1**	
	10 to 14	**B2**	Dep on correct tangent or close attempt at tangent at $x = 2$ **M1** for rise/run **also dep on** correct tangent drawn or close attempt at tangent Must see correct or implied calculation from a drawn tangent
3(e)	6	**1**	
4(a)	36.8 or 36.84…	**2**	**M1** for $\frac{h}{107} = \tan 19$ or $\frac{h}{\sin 19} = \frac{107}{\sin 71}$ oe or better
4(b)	42.1 or 42.12… from cosine rule	**4**	**M2** for $[\cos BAC =]\frac{158^2 + 132^2 - 107^2}{2 \times 158 \times 132}$ or **M1** for implicit version **A1** for $[\cos BAC =]\frac{30939}{41712}$ or 0.7417…
4(c)	35.8 or 35.84… from sine rule	**3**	**M2** for $\frac{86 \times \sin 116}{132}[= 0.58557...]$ or **M1** for $\frac{\sin CAD}{86} = \frac{\sin 116}{132}$ oe
4(d)	9670 or 9669 to 9676	**3**	**M2** for $\frac{1}{2} \times 158 \times 132 \times \sin(\textit{their}\ (\text{b}))$ oe and $\frac{1}{2} \times 86 \times 132 \times \sin(64 - \textit{their}\ (\text{c}))$ oe or **M1** for either area
4(e)	214.2 or 214.1… or 214	**2**	**M1** for [180 +]70–*their* (c) oe
5(a)(i)	52	**1**	
5(a)(ii)	36	**1**	
5(a)(iii)	26	**1**	**FT** 62 – *their* (a)(ii) evaluated correctly
5(b)	Valid comment	**1**	Strict **FT** *their* (a)(iii), e.g. distances for females are more varied
5(c)	$\frac{11}{20}$ oe	**2**	**M1** for 27 written or answer of $\frac{27}{60}$ oe
5(d)(i)	[18 9] 14 12 5 [2]	**2**	**B1** for 1 correct value

Question	Answer	Marks	Partial Marks
5(d)(ii)	48.75 nfww	**4**	**M1** for midpoints soi **M1** for use of $\sum fx$ with *their* frequencies **M1** (dep on 2nd M1) for $\sum fx \div$ (60 or by *their* $\sum f$)
6(a)(i)	Angle *ABC*=52 nfww	**B1**	ALTERNATIVE [Reflex] angle *AOC* = 256
	Opposite angles in cyclic quad oe Angles in opposite segments	**B1**	Angle at centre=2 × angle at circumference/arc
	[Angle *AOC*=104] Angle at centre=2 × angle at circumference/arc nfww	**B1**	Angles around a point
6(a)(ii)	22 nfww	**2**	**B1** for angle *OAC* = 38 or angle *CAD* = 24
6(a)(iii)	28	**1**	
6(a)(iv)	36.6 or 36.62 to 36.63 nfww	**3**	**B2** for 7.4 or 17.42 to 17.43 or **M2** for $9.6\times 2+\frac{104}{360}\times 2\times\pi\times 9.6$ or **M1** for $\frac{104}{360}\times 2\times\pi\times 9.6$
6(b)(i)	81	**3**	**M2** for $\frac{A}{36}=\left(\sqrt[3]{\frac{2187}{648}}\right)$ oe or better or for $A\times\frac{648}{36}\times\sqrt[3]{\frac{2187}{648}}=2187$ oe or better or **M1** for $\frac{A^3}{36^3}=\frac{2187^2}{648^2}$ oe or $\sqrt[3]{\frac{2187}{648}}$ or $\sqrt[3]{\frac{648}{2187}}$
6(b)(ii)	8.05 or 8.051 to 8.052…	**3**	**M2** for $\left[r^3=\right]\frac{2187\times 3}{4\times\pi}$ oe or **M1** for $\frac{4\pi r^3}{3}=2187$ **SC2** for $\frac{648\times 3}{4\times\pi}$ or **SC1** for $\frac{4\pi r^3}{3}=648$
7(a)	Reflection $y=-1$	**2**	**B1** for each
7(b)(i)	Image at (–6, 5) (–6, 7) (–5, 7) (–4, 5)	**2**	**B1** for translation by $\begin{pmatrix}-3\\k\end{pmatrix}$ or $\begin{pmatrix}k\\4\end{pmatrix}$